

Molecular Dynamics Simulations in 2D

Computational Physics Report

Tomasz Karpiński

December 20, 2025

1 Introduction

The objective of this project is to simulate the dynamics of a system of N argon atoms confined in a two-dimensional square box of width L . The simulation aims to investigate the properties of the system across different states (gas, liquid, solid) by observing the motion of particles governed by interatomic forces.

The interactions between atoms are modeled using the Lennard-Jones (LJ) potential, which describes the attractive and repulsive forces between neutral atoms. The system evolves in time according to Newton's equations of motion, which are solved numerically. To minimize finite size effects and mimic an infinite system, Periodic Boundary Conditions (PBC) are applied. The simulation uses effective (reduced) units to simplify calculations, where mass m , Boltzmann constant k_B , and LJ parameters ϵ and σ are set to unity ($m = 1, k_B = 1, \epsilon = 1, \sigma = 1$).

2 Numerical Method

2.1 Lennard-Jones Potential and Forces

The dynamics of particles are governed by interparticle interactions modeled by the shifted Lennard-Jones potential. Each atom interacts only with neighbors within a cutoff radius r_{cut} . The standard LJ potential is defined as:

$$U_{LJ}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] \quad (1)$$

To ensure the potential goes smoothly to zero at the cutoff, a shifted potential U_{LJ}^{shift} is used:

$$U_{LJ}^{shift}(r_{ij}) = U_{LJ}(r_{ij}) - U_{LJ}(r_{cut}) - \left. \frac{\partial U_{LJ}(r_{ij})}{\partial r_{ij}} \right|_{r_{ij}=r_{cut}} \cdot (r_{ij} - r_{cut}) \quad (2)$$

The force exerted on particle i by particle j is calculated using the chain rule $\vec{F}_{ij} = -\nabla_{\vec{r}_i} U_{LJ}^{shift}(r_{ij})$:

$$\vec{F}_{ij} = -\frac{\partial U_{LJ}^{shift}(r_{ij})}{\partial r_{ij}} \nabla_{\vec{r}_i} r_{ij} \quad (3)$$

where the scalar derivative is:

$$\frac{\partial U_{LJ}^{shift}(r_{ij})}{\partial r_{ij}} = 4\epsilon \left[-\frac{12\sigma^{12}}{r_{ij}^{13}} + \frac{6\sigma^6}{r_{ij}^7} \right] - 4\epsilon \left[-\frac{12\sigma^{12}}{r_{cut}^{13}} + \frac{6\sigma^6}{r_{cut}^7} \right] \quad (4)$$

2.1.1 Periodic Boundary Conditions (PBC)

To calculate the effective distance r_{ij} in a periodic box, we define parameters α and β to handle the minimum image convention:

$$\alpha = \begin{cases} -1, & x_j - x_i > L/2 + 1, \\ x_j - x_i \leq -L/2, & \text{otherwise} \end{cases}, \quad \beta = \begin{cases} -1, & y_j - y_i > L/2 + 1, \\ y_j - y_i \leq -L/2, & \text{otherwise} \end{cases} \quad (5)$$

The real distance is then:

$$r_{ij} = \sqrt{(x_j - x_i + \alpha L)^2 + (y_j - y_i + \beta L)^2} \quad (6)$$

Interactions are calculated only if $r_{ij} < r_{cut}$.

2.2 Time Iteration: Verlet Method and Thermalization

The equations of motion are solved using the Velocity Verlet method combined with a Nose-Hoover thermostat to maintain the target temperature T . The thermostat introduces a friction variable ξ and a coupling parameter Q . The time integration over a step Δt proceeds as follows:

1. First Half-Step (Velocities): Velocities are predicted using current forces and the friction variable ξ :

$$v_{xp,i} = v_{x,i}(t) + \frac{\Delta t}{2}(a_{x,i}(t) - \xi \cdot v_{x,i}(t)) \quad (7)$$

$$v_{yp,i} = v_{y,i}(t) + \frac{\Delta t}{2}(a_{y,i}(t) - \xi \cdot v_{y,i}(t)) \quad (8)$$

2. Position Update: Positions are updated using the predicted partial velocities:

$$x_i(t + \Delta t) = x_i(t) + v_{xp,i}\Delta t, \quad y_i(t + \Delta t) = y_i(t) + v_{yp,i}\Delta t \quad (9)$$

PBC are applied immediately to keep particles within $[0, L]$.

3. Force Calculation: New forces and accelerations $\vec{a}(t + \Delta t)$ are calculated using the new positions.

4. Second Half-Step (Thermostat and Velocity Correction): The new friction variable ξ and final velocities are found iteratively (Picard iteration) because the equations are implicit. For $k = 1 \dots k_{max}$:

1. Calculate instantaneous kinetic energy E_{kin} and temperature $\mathcal{T}$.

2. Update ξ :

$$\xi_{new} = \xi_{old} + \frac{\Delta t}{Q}(E_{kin} - T(N + 1)) \quad (10)$$

3. Update velocities:

$$v_{x,i}(t + \Delta t) = \frac{1}{1 + \frac{\Delta t}{2}\xi_{new}} \left(v_{xp,i} + \frac{\Delta t}{2}a_{x,i}(t + \Delta t) \right) \quad (11)$$

3 Results

3.1 Task 1: Initial Conditions

The simulation is initialized with particles placed on a regular lattice to avoid overlaps that would result in excessively high potential energy. Initial velocities are assigned from a Maxwell-Boltzmann distribution and adjusted to remove center-of-mass drift.

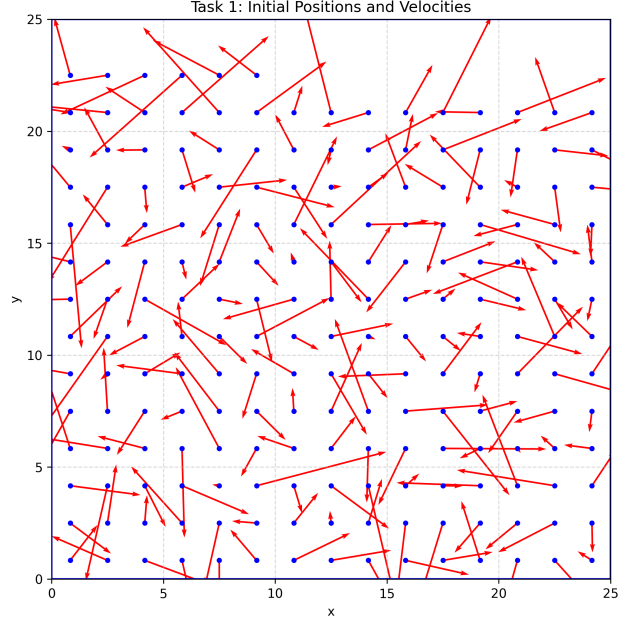
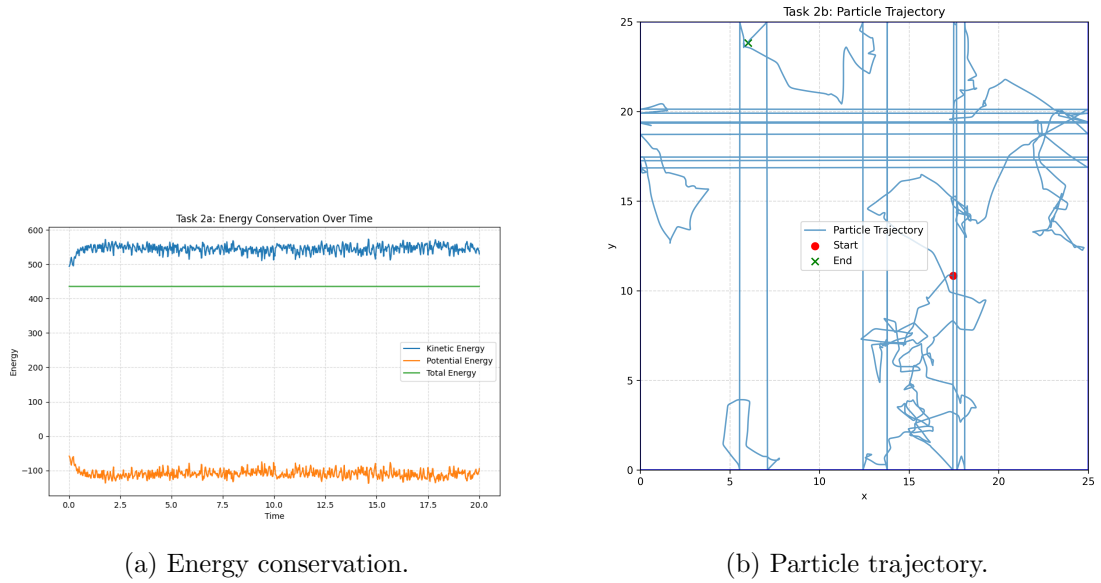


Figure 1: Initial positions of $N = 201$ atoms on a regular grid. Vectors indicate initial random velocities.

Fig. 1 shows the regular arrangement of particles. This lattice structure melts quickly as the simulation proceeds and the system thermalizes.

3.2 Task 2: Energy Conservation Trajectory

We verified the stability of the Velocity Verlet integrator by running a microcanonical simulation (thermostat off, $Q \rightarrow \infty$).



(a) Energy conservation.

(b) Particle trajectory.

Figure 2: (a) Time evolution of kinetic, potential, and total energy in the microcanonical ensemble. (b) Trajectory of a single particle undergoing diffusive motion.

As shown in Fig. 2(a), the total energy remains constant (green line) while kinetic and potential energies fluctuate, confirming the correctness of the integration scheme. Fig. 2(b) illustrates the chaotic, random-walk-like trajectory of a particle due to collisions.

3.3 Task 3: Thermalization

We investigated the effect of the thermostat coupling parameter Q on the thermalization process.

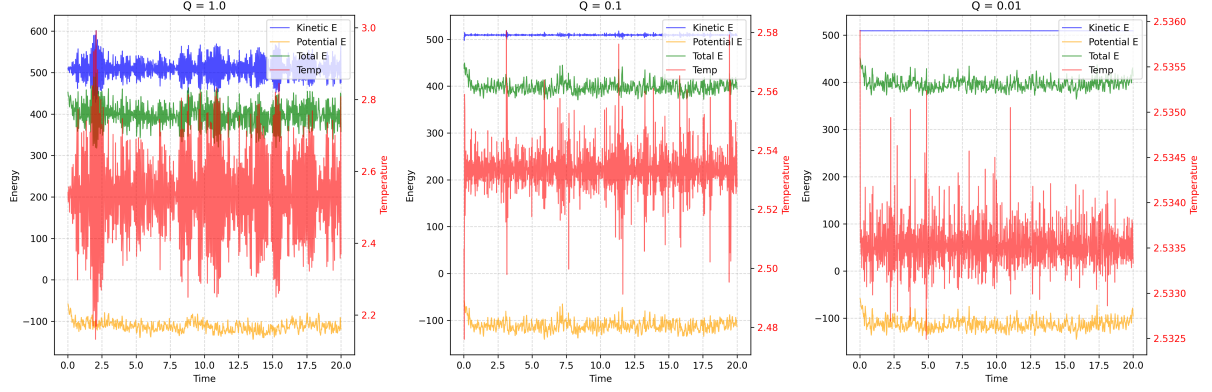


Figure 3: Energy and temperature fluctuations for different thermostat coupling parameters Q . Left: $Q = 1.0$ (weak coupling), Center: $Q = 0.1$, Right: $Q = 0.01$ (strong coupling).

Fig. 3 demonstrates that smaller Q values lead to tighter control of the temperature (smaller fluctuations around the target), effectively modeling a stronger coupling to the heat bath. For $Q = 0.01$, the fluctuations are minimal.

3.4 Task 4: Velocity Distribution

The velocity distribution of the gas was accumulated over simulations of varying lengths ($N_{it} = 10^3, 10^4, 10^5$) and compared to the theoretical Maxwell-Boltzmann distribution for 2D gas:

$$f(v) = \frac{m}{k_B T} v \exp\left(-\frac{mv^2}{2k_B T}\right) \quad (12)$$

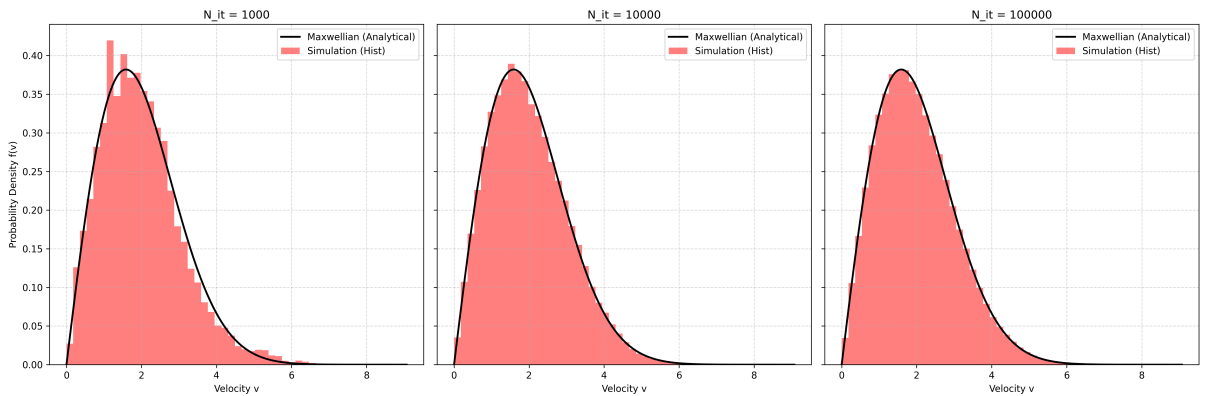


Figure 4: Velocity histograms compared with the analytical Maxwellian distribution (black line) for different simulation durations.

Fig. 4 shows that as the simulation time increases (N_{it} increases), the collected histogram converges to the theoretical analytical curve. $N_{it} = 10^5$ provides excellent agreement.

3.5 Task 5: Crystallization

The system was cooled from $T = 300$ K to $T = 5$ K using two different cooling rates: fast ($N_{it} = 10^4$) and slow ($N_{it} = 10^5$).

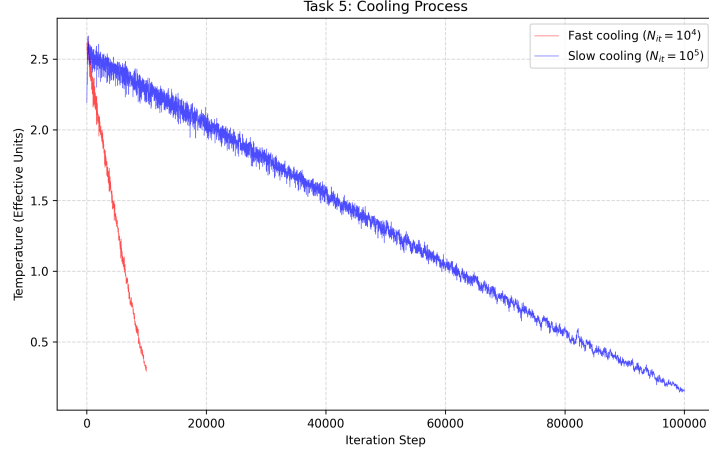


Figure 5: Instantaneous temperature during the cooling process for fast and slow cooling schedules.

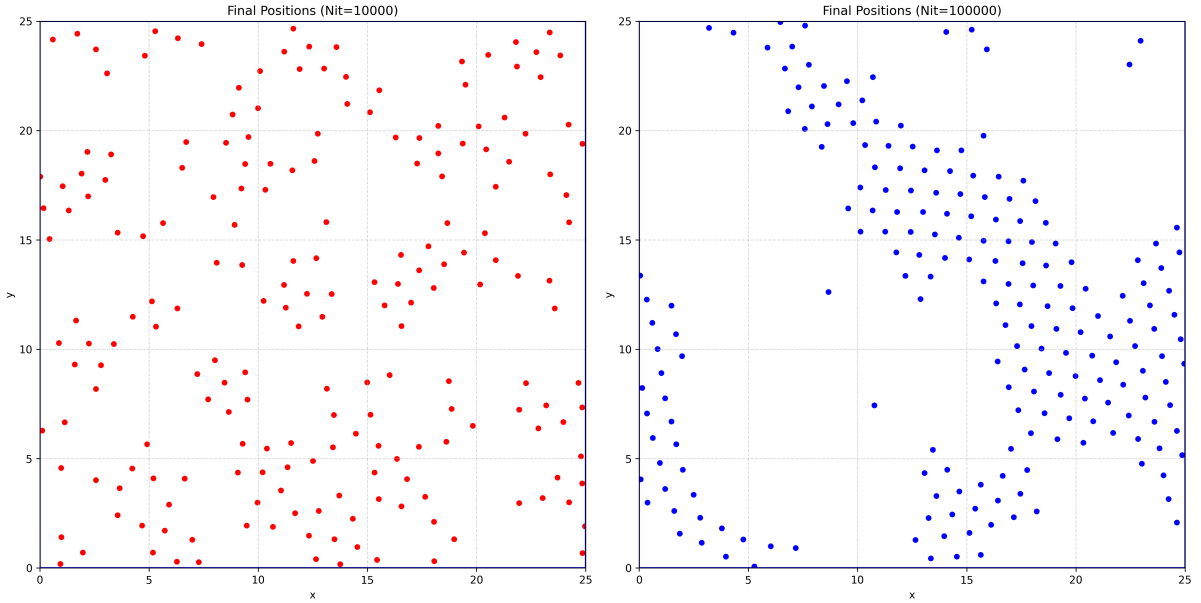


Figure 6: Final atomic positions after cooling. Left: Fast cooling (Red). Right: Slow cooling (Blue). Slow cooling allows the system to find a more ordered, hexagonal-like crystal structure.

Fig. 5 shows the stepwise decrease in temperature. Fig. 6 reveals that slow cooling (blue dots) results in a more ordered, hexagonal-like crystal structure, whereas fast cooling (red dots) freezes the system in a more disordered, glass-like state with defects (grain boundaries).

4 Conclusions

In this project, we successfully implemented a 2D Molecular Dynamics simulation using the Velocity Verlet algorithm and a Lennard-Jones potential.

- **Stability:** The integrator proved stable, conserving total energy in the microcanonical ensemble (Task 2).
- **Thermostat:** The implementation of the thermostat allowed for effective temperature control, with the coupling parameter Q regulating the magnitude of fluctuations (Task 3).
- **Statistical Mechanics:** The particle velocities naturally relaxed to the Maxwell-Boltzmann distribution, confirming the ergodic hypothesis for this system (Task 4).
- **Phase Transition:** The cooling experiment (Task 5) highlighted the importance of cooling rate in crystallization. Slow cooling allowed atoms enough time to rearrange into the energetically favorable hexagonal lattice, while rapid quenching trapped them in local minima, resulting in an amorphous or poly-crystalline structure.

The simulation effectively demonstrates fundamental concepts of statistical physics and the capability of MD to model phase transitions and material properties.